

RATIO, PROPORTION & PARTNERSHIP

THEORY

Ratio

- A ratio is a fraction (either proper or improper) which compares two or more quantities of similar kind, which enables us to understand as to how many times one quantity is involved in the other.
- If $A : B$ ($\frac{A}{B}$) is a ratio, then the numerator A is called “**Antecedent**” and the denominator B is called the “**Consequent**”.
- Ratios must be expressed in the simplest possible form and we can calculate ratios only when the quantities are commensurable (fully quantifiable).
- Two or more ratios can be bridged in order to have a continuous comparison between more than two variables.
- Rule for bridging more than two ratios :

If ,a,b,c,d,e are five Quantities, and

$$\frac{a}{b} = \frac{N_1}{D_1}, \frac{b}{c} = \frac{N_2}{D_2}, \frac{c}{d} = \frac{N_3}{D_3}, \frac{d}{e} = \frac{N_4}{D_4}$$

Then, $a:b:c:d:e = N_1N_2N_3N_4 : D_1N_2N_3N_4 : D_1D_2N_3N_4 : D_1D_2D_3N_4 : D_1D_2D_3D_4$

Let $a : b$ is a ratio, then:

- $\frac{a}{b} > 1$ (Ratio of Greater Inequality)
- $\frac{a}{b} < 1$ (Ratio of Lesser Inequality)
- $\frac{a}{b} = 1$ (Ratio of Equality)

- $a^2 : b^2$ (Duplicate Ratio)
- $a^3 : b^3$ (Triplicate Ratio)
- $\sqrt{a} : \sqrt{b}$ (Sub-Duplicate Ratio)
- $\sqrt[3]{a} : \sqrt[3]{b}$ (Sub-Triplicate Ratio)
- $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} = \dots$ If then the value of each ratio can be obtained by mean of any one of the following two operations;
 - Each ratio = $\frac{a + c + e + \dots}{b + d + f + \dots}$ (ADDENDO)
Or
 - Each ratio = $\frac{a - c - e - \dots}{b - d - f - \dots}$ (SUBTRANDENDO)

INVERSE RATIO:

- IR of a:b is b : a
- IR of a:b:c is bc : ac : ab
- IR of a:b:c:d is bcd : acd : abd : abc

COMPOUND RATIO:

The multiplying effect of all ratios given is known as compound ratio. If a:b and c:d are two ratios, then ac : bd is called the compounded ratio of the two.

Proportion

- Proportion is defined as the equality of two or more ratios. If $\frac{a}{b} = \frac{c}{d}$, in such a case the quantities a,b,c,d are said to be proportional, here 'd' is called the fourth proportional.
- If $\frac{a}{b} = \frac{b}{c}$, then a,b,c are said to be in continued proportion, where 'b' is called the mean proportional and 'c' is called third proportional.
- If $\frac{a}{b} = \frac{b}{c}$ or $b^2 = ac \therefore b = \sqrt{ac}$

IF	THEN	PROPERTY
$\frac{a}{b} = \frac{c}{d}$	$ad = bc$	PRODUCT OF EXTREMES = PRODUCT OF MEANS
	$\frac{b}{a} = \frac{d}{c}$	INVERTENDO
	$\frac{a}{c} = \frac{b}{d}$	ALTERNENDO
	$\frac{a+b}{b} = \frac{c+d}{d}$	COMPONENDO
	$\frac{a-b}{b} = \frac{c-d}{d}$	DIVIDENDO
	$\frac{a+b}{a-b} = \frac{c+d}{c-d}$	COMPONENDO & DIVIDENDO
	$\frac{a-b}{a+b} = \frac{c-d}{c+d}$	DIVIDENDO & COMPONENDO